$\Theta \rightarrow$ Distance Matric

Given

$$\mathcal{D} = \{x_1, x_2, \dots, x_N\}. \sim \text{iid } \mathbb{P}_x$$

(unknown).

Goal: Estimate $\mathbb{P}_x$ and learn to sample from it.

General principle of Gen. Models:

(i) Assume the parametric family on P_x, denoted by P_θ.

~~P_θ: Repeated~~ using

P_θ: Represented using Deep Neural Network (Model)

(ii) Define & estimate a divergence (distance) metric between P_θ & P_x.

(iii) Solve an optimization problem over the parameters of P_θ, to minimize the above divergence Metric

(iv)

Example:-

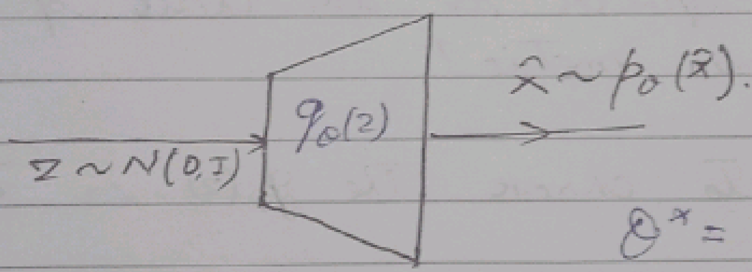
There is a R.V $z \in \mathbb{R}^k$ and has some arbitrary but known distribution

$$z \sim N(0, I)$$

Suppose $g_\theta(z) : Z \rightarrow X$, Then

$\hat{x} = g_\theta(z)$ has a different distribution than that of z & the distribution of $\hat{x}$; depends of the function $g_\theta()$.

Suppose $g_\theta(z)$ is a Neural Network. Denote the density of $\hat{x} = g_\theta(z)$. using $p_\theta(\hat{x})$.



$$\theta^* = \underset{\theta}{\operatorname{argmin}}\ D(p_x \| p_\theta)$$

Suppose $D(p_x \| p_\theta)$ denote a divergence measure between p_x & p_θ, $D \geq 0$, $D = 0$. iff $p_x = p_\theta$.

Inference.

A sample from $z \sim \mathcal{N}(0, I)$, passed through $g_{\theta^*}(z)$ would produce a sample from $p_{\theta^*}(x)$ which is close to p_x, we end up.

Sampling from p_x.

The Questions :-

i) How ~~do~~ to compute the divergence metrics without knowing p_x & p_θ?

↳ Since we just have samples from p_x & p_θ not the actual distribution how are we going to compute the divergence metrics.

ii) What should be the choice of the divergence metric?

iii) How to choose the $g_\theta(z)$, in turn p_θ?

iv) How to solve the optimization problem of minimizing the divergence metrics?

Variational Divergence Minimization

Define divergence metrics between distributions

f-Divergence

Given two probability distribution functions with the corresponding density functions denoted by p_x, & p_θ, the f-divergence between them is defined as follows:

$$\cancel{\boxed{D_f(p_x \| p_\theta) = \int_X p_x(x)\, f\left(\frac{p_x(x)}{p_\theta(x)}\right) dx}}$$

$$D_f(p_x \| p_\theta) = \int_X p_\theta(x)\, f\left(\frac{p_x(x)}{p_\theta(x)}\right) dx.$$

$f(u) : \mathbb{R}^+ \rightarrow \mathbb{R}$, convex, left-semi continuous
$f(1) = 0$.

X: space on which the p_x & p_θ are supported.

properties of f-Divergence:-

i) $D_f(\,) \geq 0$ for any choice of $f(\,)$.

ii) $D_f(p_x \| p_\theta) = 0$ iff $p_x = p_\theta$.

a function $f(u)$ is convex iff $\forall u_1, u_2 \in U$ &
$0 \leq \alpha_1, \alpha_2 \leq 1$, $\alpha_1 f(u_1) + \alpha_2 f(u_2) \geq f(\alpha_1 u_1 + \alpha_2 u_2)$

Examples of f-divergence:

a) $f(u) = u \log u$, KL - divergence

$$D_f(p_x \| p_\theta) = \int_X p_\theta(x) f\left(\frac{p_x(x)}{p_\theta(x)}\right) dx$$

$$= \int_X p_\theta(x) \times \left[\frac{p_x(x)}{p_\theta(x)} \times \log\left(\frac{p_x(x)}{p_\theta(x)}\right)\right] dx$$

$$= \int_X p_x(x) \log \frac{p_x(x)}{p_\theta(x)} dx = D_{KL}.$$

$$\underbrace{D_{KL}(p_x \| p_\theta)}_{\text{forward KL}} \neq \underbrace{D_{KL}(p_\theta \| p_x)}_{\text{reverse KL.}}$$

ii) JS- divergence

$$f(u) = \frac{1}{2}\left(u \log u - (u+1) \log\left(\frac{u+1}{2}\right)\right)$$

iii) $f(u) = \frac{1}{2}|u-1|$: Total variation distance

⊙→ Algorithm for D_f minimization

⊙ Inference of Divergence Metric.

Ex:- $D_{KL} = \int p_x \log \frac{p_x(x)}{p_\theta(x)} dx$

Consider $\hat{x}$ where $p_x(\hat{x}) \uparrow$ $p_\theta(\hat{x}) \downarrow$.
↳ Here D_{KL} is high ~~and~~ and it is a desired behaviour.
↳ Model is not generating what has to generated.

$\hat{x}$ when $p_x(\hat{x}) \downarrow$ & $p_\theta(\hat{x}) \uparrow$.
↳ Here D_{KL} is low which is not desirable.
↳ Here it ~~is~~ is generating junk.